Gravitation

1. Two-point masses A and B having masses in the ratio 4 : 3 are separated by a distance of 1 m. When another point mass C of mass M in is placed in between A and B, the force between A and C is $\frac{1}{3}rd$ of the force between B and C. Then the distance of C from A is

1) $\frac{2}{3}m$
2) $\frac{1}{3}m$
3) $\frac{1}{4}m$
4) $\frac{2}{5}m$

2. Two sphere of mass m and M are situated in air and the gravitational force between them is F. The space around the masses is now filled with a liquid of specific gravity 3. The gravitational force will now be

1) F
2) $\frac{F}{3}$
3) $\frac{F}{9}$
4) $3F$

3. Three equal masses of 1 kg each are placed at the vertices of an equilateral triangle PQR and a mass of 2 kg is placed at the centroid O of the triangle which is at a distance of $\sqrt{2}$ m from each of the vertices of the triangle. The force, in newton, acting on the, mass of 2 kg is

1) 2
2) $\sqrt{2}$
3) 1
4) Zero

4. A body of mass 500 g is thrown upward with a velocity 20 m/s and reaches back to the surface of a planet after 20 sec Then the weight of the body on that planet is

1) 2 N
2) 4 N
3) 5 N
4) 1 N

5. If g is the acceleration due to gravity on the surface of earth, its value at a height from the surface equal to double to radius of earth is

1) g
2) g/2

3) g/3
4) g/9

6. A body weighs 700 g wt on the surface of the earth. How much will it weigh on the surface of a planet whose mass is $\frac{1}{7}$ and radius is half that of the earth

1) 200 g wt
2) 400 g wt
3) 50 g wt
4) 300 g wt

7. The masses of two planets are in the ratio 1: 2. Their radii are in the ratio 1: 2. The acceleration due to gravity on the planets are in the ratio

1) $1:2$
2) $2:1$
3) $3:5$
4) $5:3$

8. The ratio between the values of acceleration due to gravity at a height 1 km above and at a depth of 1 km below the earth's surface is (radius of earth is R)

1) $\frac{R-2}{R-1}$
2) $\frac{R}{R-1}$
3) $\frac{R-2}{R}$
4) $\frac{2R}{R-1}$

9. The depth d at which the value of acceleration due to gravity becomes 1/n times the value at the surface, is [R = radius of the earth]

1) $\frac{R}{n}$
2) $R\left(\frac{n-1}{n}\right)$
3) $\frac{R}{n^2}$
4) $R\left(\frac{n}{n+1}\right)$

10. Acceleration due to gravity on moon is 1/6 of the acceleration due to gravity on earth. If the ratio of densities of earth (ρ_e) and moon (ρ_m) is $\left(\frac{\rho_e}{\rho_m}\right)=\frac{5}{3}$ then radius of moon R_m in terms of R_e will be

1) $\frac{5}{18}R_e$

2) $\frac{1}{6}R_e$

3) $\frac{3}{18}R_e$

4) $\frac{1}{2\sqrt{3}}R_e$

11. A body of mass 'm' is taken from the earth's surface to the height equal to twice the radius (R) of the earth. The change in potential energy of body will be

1) $\frac{1}{3}mgR$

2) $mg2R$

3) $\frac{2}{3}mgR$

4) $3mgR$

12. The mass of spaceship is 1000 kg. It is to be launched from the earth's surface out into free space. The value of 'g' and R' (radius of earth) are 10 m/s² and 6400 km respectively. The required energy for this work will be

1) 6.4×10^{11} Joules

2) 6.4×10^{8} Joules

3) 6.4×10^{9} Joules

4) 6.4×10^{10} Joules

13. A particle of mass M is situated at the centre of a spherical shell of same mass and radius a. The magnitude of the gravitational potential at a point situated at a/2 distance from the centre, will be

1) $\frac{4GM}{a}$

2) $\frac{GM}{a}$

3) $\frac{2GM}{a}$

4) $\frac{3GM}{a}$

14. Two bodies of masses m and 4m are placed at a distance r. The graviatonal potential at a point on the line joining them where the gravitational field is zero is

1) Zero

2) $-\frac{4GM}{r}$

3) $-\frac{6GM}{r}$

4) $-\frac{9GM}{r}$

15. Two small and heavy spheres, each of mass M, are placed a distance r apart on a horizontal surface. The gravitational potential at the mid-point on the line joining the center of the sphere of

1) Zero
2) $-\frac{GM}{r}$
3) $-\frac{2GM}{r}$
4) $-\frac{4GM}{r}$

16. 3 particles each of mass m are kept at vertices of an equilateral triangle of side L. The gravitational field at centre due to these particles is

1) Zero
2) $\frac{3GM}{L^2}$
3) $\frac{9GM}{L^2}$
4) $\frac{12}{\sqrt{3}}\frac{GM}{L^2}$

17. The potential energy of 4-particles each of mass 1 kg placed at the four vertices of a square of side length 1 m is

1) $+4.0\,\mathrm{G}$
2) $-7.5\,\mathrm{G}$
3) $-5.4\,\mathrm{G}$
4) $+6.3\,\mathrm{G}$

18. Energy required to move a body of mass m from an orbit of radius 2R to 3R is

1) GMm/12R
2) GMm/3R
3) GMm/8R
4) GMm/6R

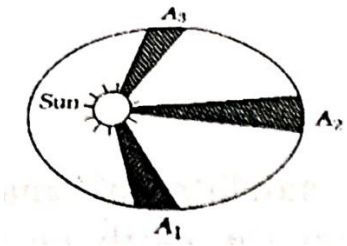
19. A point mass is placed inside a thin spherical shell of radius R and mass M at a distance R/2 from the centre of the shell, The gravitational force exerted by the shell on the point mass is

1) Zero
2) $-\frac{4Gm}{r}$
3) $-\frac{6Gm}{r}$
4) $-\frac{9Gm}{r}$

20. Infinite number of masses, each 1 kg, are placed along the x-axis at $x = \pm 1m, \pm 2m, \pm 4m,,, \pm 8m, \pm 16m....$ the magnitude of the resultant gravitational potential in terms of gravitational constant G at the origin $(x = 0)$ is

1) $\frac{G}{2}$
2) G
3) $2G$
4) $4G$

21. A planet moving around sun sweeps area A_1 in 2 days, A_2 in 3 days and A_3 in 6 days. Then the relation between A_1, A_2 and A_3 is

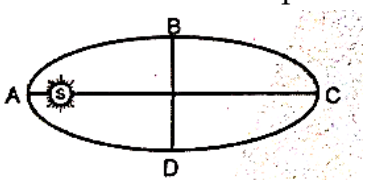


1) $3A_1 = 2A_2 = A_3$
2) $2A_1 = 3A_2 = 6A_3$
3) $3A_1 = 2A_2 = 6A_3$
4) $6A_1 = 3A_2 = 2A_3$

22. A planet moves around the sun. At a given point P, it is closet from the sun at a distance d_1 and has a speed v_1. At another point Q, when it is farthest from the sun at a distance. d_2, its speed will be

1) $\frac{d_2^2 v_1}{d_2^2}$
2) $\frac{d_2 v_1}{d_1}$
3) $\frac{d_1 v_1}{d_2}$
4) $\frac{d_2^2 v_1}{d_1^2}$

23. The earth E moves in an elliptical orbit with the sun S at one of the foci as shown in figure. Its speed of motion will be maximum at the point



1) C
2) A
3) B
4) D

24. If the height of satellite from the earth is negligible in comparison to the radius of the earth R, the orbital velocity of the satellite is

1) gR
2) $gR/2$
3) $\sqrt{g/R}$
4) $\sqrt{gR}$

25. Two satellite A and B go round a planet in circular orbits having radii 4R and R, respectively. If the speed of satellite A is 3v. then speed of satellite B is

1) $\frac{3v}{2}$
2) $\frac{4v}{2}$
3) $6v$
4) $12v$

26. The escape velocity of a body from earth's surface is v_e. The escape velocity of the same body from a height equal to 7R from earth's surface will be

1) $\frac{v_e}{\sqrt{2}}$
2) $\frac{v_e}{2}$
3) $\frac{v_e}{2\sqrt{2}}$
4) $\frac{v_e}{4}$

27. The Earth is assumed to be a sphere of radius R. A platform is arranged at a height R from the surface of the Earth. The escape velocity of body from this platform if fv, where v is its escape velocity from the surface of the Earth. The value of f is

1) $\frac{1}{3}$
2) $\frac{1}{2}$
3) $\sqrt{2}$
4) $\frac{1}{\sqrt{2}}$

28. A satellite moves around the earth in a circular orbit of radius r with speed v. If the mass of the satellite is M, its total energy is

1) $-\frac{1}{2}Mv^2$
2) $\frac{1}{2}Mv^2$

3) $\frac{3}{2}Mv^2$

4) Mv^2

29. A satellite moves round the earth in a circular orbit of radius R making one revolution per day. A second satellite moving in a circular orbit, moves round the earth one in 8 days. The radius of the orbit of the second satellite is

1) 8R
2) 4R
3) 2R
4) R

30. The value of escape velocity on a certain planet is 2 km/s. Then the value of orbital speed for a satellite orbiting close to its surface is

1) 12 km/s
2) 1 km/s
3) $\sqrt{2}$ km/s
4) $2\sqrt{2}$ km/s

Answer Key

(1)	1	**(16)**	1
(2)	1	**(17)**	3
(3)	4	**(18)**	4
(4)	4	**(19)**	1
(5)	4	**(20)**	4
(6)	2	**(21)**	1
(7)	2	**(22)**	3
(8)	1	**(23)**	2
(9)	2	**(24)**	4
(10)	1	**(25)**	3
(11)	3	**(26)**	3
(12)	4	**(27)**	4
(13)	4	**(28)**	1
(14)	4	**(29)**	2
(15)	4	**(30)**	3